



Case Report

Adjuvant multisite bacteriophage cocktail administration as salvage treatment in a patient with a multidrug-resistant *Pseudomonas aeruginosa* frontal relapsing complex osteitis

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ABSTRACT

Background: Bone infection due to multidrug-resistant (MDR) *Pseudomonas aeruginosa* is associated with poor outcomes, especially high risk of relapses. Adjunctive multisite phage therapy can be used as a salvage treatment in relapsing cases.

Case presentation: We treated a 37-year-old man with MDR *P. aeruginosa* frontal osteitis and a large frontal fistula, who experienced three relapses, using personalized multisite phage therapy combined with surgery and prolonged antibiotics (ceftolozane/tazobactam). The multisite phage administration included *in situ* instillation of the manufactured phage cocktail during surgery, followed by daily intravenous infusion and nebulization for 7 days after surgery. No adverse events were reported during treatment. At the 2-year follow-up, the patient had no relapse without any suppressive treatment.

Conclusion: Multisite phage administration is a personalized treatment strategy that can prevent relapse or the emergence of additional antimicrobial resistance in complex cases, such as relapsing MDR *P. aeruginosa* osteitis.

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Introduction

Bone and joint infections (BJI) are difficult-to-treat infections and are associated with high relapse rates due to biofilm formation. This is even more challenging when BJI involves multidrug-resistant (MDR) bacteria, such as *Pseudomonas aeruginosa*. Virulent bacteriophages (phages for short) are natural viruses that selectively target, replicate inside, and lyse planktonic and biofilm-forming bacteria *in vitro* [1]. In addition, some phages act against

the biofilm, whereas few antimicrobials are effective against this bacterial-produced matrix [2].

In the specific setting of BJI due to MDR *P. aeruginosa*, the use of adjuvant lytic phage(s) could represent a promising therapy, in combination with antibiotics and surgery [3]. However, cases of successful treatment of *P. aeruginosa* BJI after phage therapy are scarce, and limited to the spine or legs [2,3]. The optimal way of administration and dose of active phages remains unclear, and appears to depend on the clinical presentation. Here, we present a complex case of MDR *P. aeruginosa* frontal osteitis, addressing the efficacy and tolerability of personalized multisite administration of phages.

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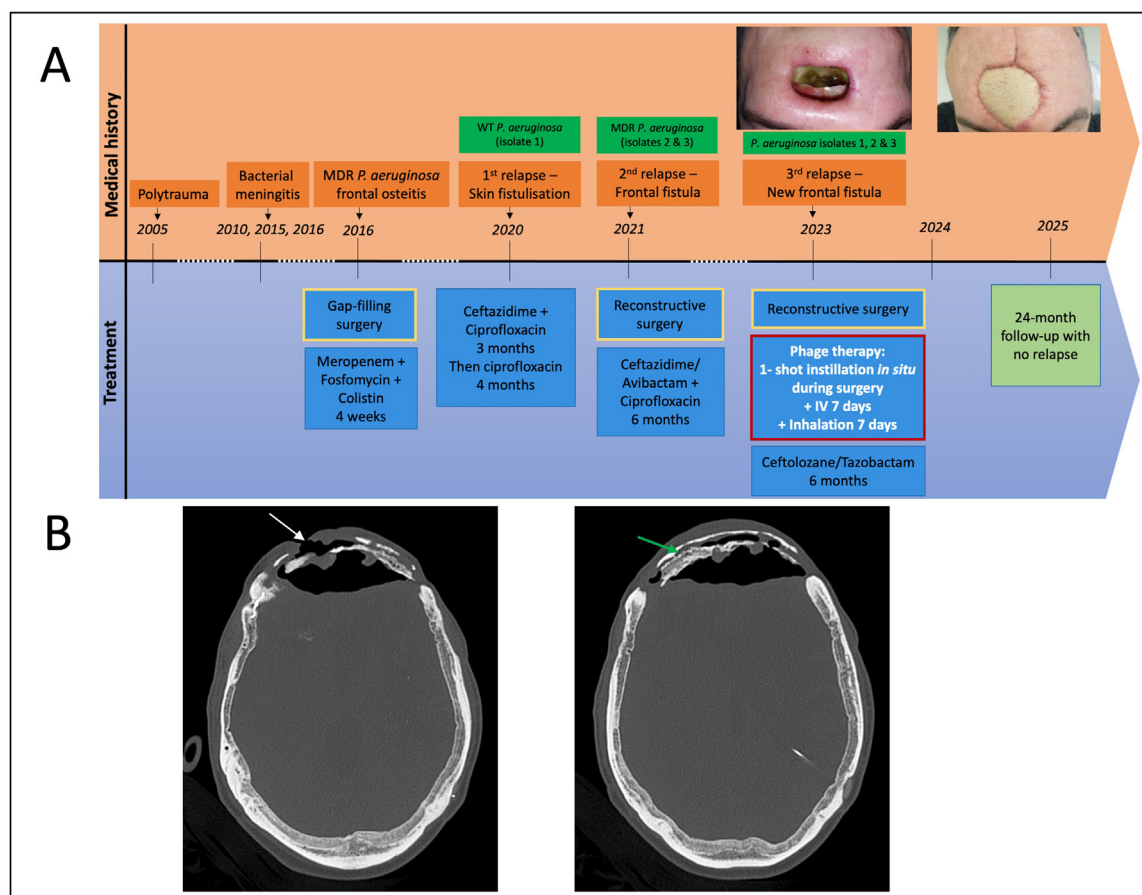


Figure 1. (A) Chronology of frontal osteitis history and management. (B) Computed tomography scan showing the frontal fistula (white arrow) and osteitis (green arrow).

Case presentation

In 2005, a 37-year-old man experienced a head injury after a 7-meter fall, resulting in an extradural haematoma, frontal and anterior skull base fractures. He subsequently developed acute and then chronic hydrocephalus requiring a ventriculo-atrial shunt. Five, ten and eleven years after the trauma, he experienced multiple episodes of presumed bacterial meningitis, for which no pathogen was identified, related to an osteo-meningeal breach. Following the gap-filling surgery, the patient presented a MDR *P. aeruginosa* frontal osteitis involving two isolates, treated with meropenem (2 g q8 h), fosfomycin (4 g q8h) and colistin (6MUI q12h) during 4 weeks. Four years later, the frontal osteitis relapsed, associated with skin fistulisation. A bone biopsy revealed a wild-type *P. aeruginosa*, and the patient was treated with a combination of ceftazidime (2 g q8h) and ciprofloxacin (750 mg b.i.d orally) for 3 months, followed by ciprofloxacin alone for 4 months.

A second relapse with frontal fistula occurred 1 year afterwards, requiring reconstructive flap surgery involving two MDR *P. aeruginosa* isolates found in bone surgical samples (Supplementary Table 1). Ceftazidime/avibactam (2 g q8h) and ciprofloxacin (750 mg b.i.d orally) were administered for 6 months, resulting in initial improvement. However, several months later, the frontal fistula reappeared again, confirming a third relapse (Figure 1A). A brain CT-scan showed a communication between the new extradural cavity and sinus below, and frontal bone lysis suggesting chronic osteitis (Figure 1B).

This case was discussed at multidisciplinary expert meetings in the referral center of complex BJI CRIOA Lyon (Hospices Civils de Lyon), and at dedicated phage therapy multidisciplinary meet-

ings throughout the “PHAGEinLYON Clinic programme” [4]. Considering the high risk of relapse, a multimodal treatment strategy was decided, involving: i) a surgical procedure of resection of the necrotic frontal bone, sinusotomy and reconstruction with a free thigh flap; ii) intravenous antibiotic treatment; iii) *in situ* phage therapy through local instillation during surgery, associated with intravenous and inhaled phage therapy for 7 days following surgery.

Three different MDR *P. aeruginosa* isolates were sent to PHAGENIX (Lyon, France) to realise phagogram, as previously described [2,5]. The phage susceptibility testing (PST) was performed on the three clinical isolates to evaluate four phages from the PHAGENIX collection [1,5]. This PST combined two complementary methods: a plaque assay (spot version) to determine the efficiency of plating score and lysis pattern; and the broth microdilution assay to determine the minimum inhibitory concentration (Supplementary Table 2).

Three bacteriophages with investigational medicinal product status, manufactured by the company in accordance with the Good Manufacturing Practice (GMP) method were selected based on their susceptibility profiles: PP1792 and PP1797 (*Bruynoghevirus LUZ24*; active against two strains each) and PP1450 (*Pbunavirus LS1*; active against one strain). Each phage was supplied as an individual medicinal product in ready-to-use 3 mL vials containing 1 mL of a 10^{10} PFU/mL phage suspension. Prior to surgery, the patient received 7 days of ceftolozane/tazobactam (2 g q8h) and 2 days of amikacin (900 mg daily).

On the day of the surgery, the hospital pharmacist aseptically withdrew 1 mL of each phage and diluted them in sterile saline to prepare a 5 mL phage cocktail solution. Plastic and ear, nose and

throat surgeons conducted a large resection of the anterior wall of the frontal bone, debrided the fistula, and milled down to the meninges and orbital floor, closing the site with an anteriolateral thigh flap, sutured to the edges of the loss of substance. The phage cocktail containing the three phages was then injected beneath the periosteum and skin, near the area of bone sequestration. Adjunctive colistin (3MUI q8h) was administered for 7 days after surgery, in combination with ceftolozane/tazobactam.

Postoperative phage therapy included daily nebulization of the phage cocktail (5 mL dilution), using a vibrating mesh-associated nebulization (Aerogen® Solo system) for 7 days. Additionally, the phage cocktail (diluted in 50 mL) was administered intravenously daily for 7 days (duration of the infusion 2–5 min), to enhance delivery to the infection site where surgery was freshly performed. No adverse events attributable to phage therapy were reported. All surgical samples were finally positive for the three previously identified strains of *P. aeruginosa*, and then ceftolozane/tazobactam monotherapy was continued for 6 months (4 g q12h) through outpatient parenteral antibiotic therapy. After 24 months of follow-up after the last surgery, no clinical or biological sign of relapse was observed.

Discussion

To the best of our knowledge, this is the first report of multi-site phage therapy for MDR *P. aeruginosa* frontal osteitis. Indeed, phage therapy is a customized treatment targeting the patient's strain, and also preventing the emergence of bacterial resistance. Combining phages that target different classes of bacterial receptors (such as LPS or efflux pumps) is a more effective strategy for limiting the emergence of resistance than using a single phage [6]. Here, personalized treatment was based on the following steps: (i) selection of a GMP-grade phage medicinal product based on their *in vitro* activity on the patient's strains; (ii) the modalities of phage administration; and (iii) the integration of phage therapy as an adjuvant medico-surgical approach.

Based on the experience acquired throughout the "PHAGEinLYON Clinic" programme, multisite administration was performed. Local treatment during the surgery is important for phage exposure, but a one-off exposure is insufficient to cover the entire infected area. To complete the phage exposure, and to take the advantage of the surgical debridement as a benefit for phage penetration (surgery induces a local bleeding in the days following the procedure), intravenous phage therapy was proposed, as it had previously been used to tackle various difficult-to-treat infections [7,8]. This was well tolerated thanks to the GMP purification process, which produced a medicinal product with a low level of endotoxin (≤ 50 EU/mL), and met the recent the general chapter of the European pharmacopeia regarding phage therapy medicinal products. Finally, given the presence of a cavity behind the infected bone and in front of the infected meninges that communicated with the upper airways to form a neosinus, we proposed inhalation of phages at the same time. The process of aerolization of phages could destroy the structure of phages, especially those belonging to the *Myoviridae* family. The nebulization parameters of these three phages were previously investigated, and the results predicted efficient dose delivery upon nebulization, particularly when using a vibrating mesh nebulizer [9].

In addition, phage anti-biofilm activity could enhance antibiotic penetration, and may improve their efficacy [10]. The three phages used in this case report showed significant antibiofilm activity *in vitro* [1]. It has been shown that phages and ceftolozane/tazobactam exhibited synergistic activity and reduced biofilm viability more effectively than monotherapy [11]. More broadly, the phage-antibiotic synergy is based on evolutionary trade-offs. Phages co-evolve with their bacterial hosts by modifying

their resistance mechanisms, such as efflux capacity and altered membrane permeability, thereby increasing susceptibility to antibiotics [12,13]. This personalized multisite administration, which was specifically set up for this case, combined with medico-surgical treatment, probably participated to the clinical success with no relapse at 2-years follow-up without any suppressive antimicrobial therapy.

Experience with phage therapy is growing, and most cases have not reported serious adverse side effects, as in the case presented here [14]. The personalization of phage therapy, combining multiple routes of administration is necessary to achieve sufficient phage exposure at the infection site in order to obtain a clinical effect. For that purpose, structuring at the national level with dedicated multidisciplinary meetings is useful, and will facilitate the performance of phage therapy clinical trials.

Data availability

Not applicable.

Consent for publication

Informed consent for publication was obtained from patient.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethics approval

Not applicable.

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Author contributions

AD, TF, MR and DLP co-wrote the original manuscript. TF led the multidisciplinary expert phage meeting that decided on multi-site phage therapy. FJ and TM performed surgery. TB, RGVB and TF performed phage susceptibility testing. AD, ALC, MR, CF and DLP were in charge of the patient. FJ, TM, TB, RGVB, CF, TF and ALC reviewed the manuscript.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ijid.2026.108756](https://doi.org/10.1016/j.ijid.2026.108756).

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